

2 — Simulation: the cloud bank

1 What the bank is

The bank is a pool of 50 000 parameter vectors θ per family, each realised as 2 patterns on $W = [0, 1]^2$: 500 000 clouds in all. Train, validation and test sets are carved from it afterwards.

The design principle is that **the bank is drawn on physical parameters and knows nothing about departure from CSR**. The departure label $\tilde{\delta}$ is applied afterwards by `scripts/relabel.py`, as a deterministic function of $(\theta, \bar{n})$ and the null tables. Refitting the tables, or adding a reduction, is a relabelling and never a re-simulation.

This inverts the earlier design, which swept $(\bar{n}, \tilde{\delta})$ directly and solved for the physical parameters. Sweeping $\tilde{\delta}$ makes the bank a hostage of the null tables: every change to the calibration invalidates every cloud. It also forces the design onto a coordinate that is not identifiable — many θ share a $\tilde{\delta}$ — so the pool silently concentrates on whichever pre-image the solver happened to find.

2 Two dials

Every structured family is drawn on two independent dials.

- **Richness** — points per cluster. Decides whether the pattern is *really* a cluster process, or a superposition of so many one-point parents that it is Poisson in disguise.
- **Smearing** — $\omega = \sigma\sqrt{\kappa}$, the cluster scale in units of the parent spacing. Decides whether you can *see* it: $\omega \ll 1$ leaves crisp clusters, $\omega \gg 1$ smears them towards CSR.

The two are independent by construction, which is what lets the bank separate “there is no structure” from “there is structure but it is invisible at this resolution”. A bank drawn on one dial alone cannot make that distinction, and it is the distinction the regime analysis rests on.

$\bar{n}$ is a budget, not a parameter. Each family draws $\bar{n} \sim \text{log-uniform}[75, 950]$, then draws its dials, and the remaining parameter falls out by conservation of the mean count: $\kappa = \bar{n}/\mu$ for Thomas, $\kappa = \bar{n}/(\mu_1\mu_2)$ for nested, and λ_p from an exact inversion of the thinning for Matérn. The realised $\bar{n}(\theta)$ reproduces its target to 2×10^{-13} .

3 Bounds are inverted, not rejected

Every bound is written into the *support* of the next draw, conditioned on what is already drawn, rather than tested after the fact. This matters more than it sounds.

A rejection is only harmless when it is uniform over the design. The constraints here are not: they concentrate almost entirely in one corner — few clusters, crisp clusters — so rejecting on them silently reweights the *richness* dial, which is the one dial that is supposed to be swept freely. Before the constraint on the count variance was inverted, nested rejected 18.2% of draws and the delivered design was tilted away from the sparse corner: $P(\kappa < 15)$ was 0.285 against an intended 0.396, with every quartile of κ , μ_1 and μ_2 displaced. Inverting the same constraint restores all of them exactly and leaves no rejections.

Two helpers do this, both solving for the boundary by `brentq` on log of the parameter:

- **cv_cap** — largest value with $cv \leq cv_{\max}$, for a parameter in which cv *increases* (LGCP field variance).
- **cv_floor** — smallest value with $cv \leq cv_{\max}$, for a parameter in which cv *decreases* (the smearing dial ω : tighter clusters concentrate the count variance). The mirror of **cv_cap**.

The constraint does not disappear, it relocates. In nested the count-variance bound now bites only on ω , per θ , on the dial it physically belongs to; the price is that ω is log-uniform on a κ -dependent feasible range rather than a fixed one. That is the honest accounting, and it is visible in the manifest.

4 The count-variance bound

One rule is shared by every family: $cv(n) = sd(n)/\bar{n} \leq cv_{\max} = 0.30$, computed in closed form from K via the isotropic set covariogram of W .

Its purpose is **not** to keep n inside the null tables' domain $[20, 2000]$ — the pattern-level filter already guarantees that, and even the worst draws the bound rejects land outside the domain only 1–3.5% of the time. Its purpose is **label integrity**. The label $\tilde{\delta}$ is evaluated at $\bar{n}$, but the pattern it labels has a realised n . The larger the cv , the further n wanders from $\bar{n}$, and the worse the label describes the cloud it is attached to. Evaluating the same curve at $n = \bar{n}(1 \pm cv)$:

cv band	median label drift, Thomas	nested
≤ 0.10	8.6%	9.1%
0.10–0.20	17.0%	17.2%
0.20–0.30	26.4%	27.1%

Drift is measured on the **ext** scale, relative to $\tilde{\delta}$, over the thetas with $\tilde{\delta} > 1$: below the boundary $\tilde{\delta}$ passes through zero and a relative drift stops being meaningful there, though the absolute drift keeps the same shape (median 0.05, 0.22, 1.66 in $\tilde{\delta}$ units for Thomas across the same three bands).

Drift is close to linear in cv , so $cv_{\max}$ is a direct dial on how much the label may lie: about a quarter of $\tilde{\delta}$ at one standard deviation at the current cap, and by extrapolation nearer 40% at the 0.45 this replaced.

5 Families and their constraints

Poisson

Homogeneous, intensity $\bar{n}$. No parameters beyond the budget, no constraints: its closed-form curve is identically zero. Under the default **ext** reduction that scores $\tilde{\delta} \approx -0.97$, the CSR noise floor, not 0 — see §8. It is the null, not a family with a vanishing amplitude.

Thomas

Parents at intensity κ , each with $\text{Poisson}(\mu)$ offspring displaced by $N(0, \sigma^2 I)$. Richness μ , smearing $\omega = \sigma\sqrt{\kappa}$; $\kappa = \bar{n}/\mu$.

- $\kappa \geq 10$ (**kappa_min**) — at least 10 clusters in the window, so κ is estimable and the pattern is a cluster process rather than one blob. Enters as the ceiling $\mu \leq \bar{n}/10$.
- $\mu \geq 1$ (**mu_min**) — a floor on points per cluster. Below 1 the process is a thinned Poisson wearing a cluster parameterisation.
- $\omega \in [0.2, 8]$ (**omega**) — the smearing dial, log-uniform.
- $\sigma \leq 1.5$ (**sigma_max**) — a *cost* guard, not a modelling choice: parents are simulated on W dilated by 5 displacement s.d.s, so the parent window grows as $(1 + 10\sigma)^2$. It caps ω at $\min(8, 1.5\sqrt{\kappa})$, which bites whenever $\kappa < 28.4$, i.e. for 33.8% of the delivered thetas. It never rejects: $\sigma \leq 1.5$ would require $\kappa \leq 0.018$ to leave no room, and $\kappa \geq 10$.

- $cv \leq 0.30$, inverted into a floor on ω by `cv_floor`.

Rejections: none. All four bounds are inverted; the infeasible branches are unreachable, not merely unused. At the top of the feasible ω range the worst cv over the whole design is 0.129, so the `cv_floor` failure path cannot fire.

Nested

The same construction at two levels: meta-parents at κ , Poisson(μ_1) parents displaced at scale σ_1 , Poisson(μ_2) children at scale σ_2 . Richness (μ_1, μ_2) , smearing $\omega = \sigma_1 \sqrt{\kappa}$; $\kappa = \bar{n}/(\mu_1 \mu_2)$.

- $\kappa \geq 10$, as above, entering twice: first as $\mu_2 \leq \bar{n}/10$, then as $\mu_1 \leq \bar{n}/(10\mu_2)$ conditioned on the μ_2 already drawn.
- $\mu_2 \geq 1$ (`mu2_min`) — children per parent.
- $\mu_1 \mu_2 \geq 2$ (`meta_min`) — at least two points per meta-cluster. Without it the outer level is a degenerate relabelling of the inner one and nested collapses to Thomas.
- $\rho = \sigma_1/\sigma_2 \in [2.49, 10]$ (`rho`) — scale separation. The companion to `meta_min`: the two levels must be distinct in *scale*, not merely in count, or the two-level K is indistinguishable from a one-level K at an intermediate σ .
- $\omega \in [0.2, 8]$ with the same $\sigma_1 \leq 1.5$ cost cap, which here bites for 70.4% of thetas because nested κ runs lower.
- $cv \leq 0.30$, inverted by `cv_floor`. This requires ρ to be drawn *before* σ_1 , since the inversion needs the complete model to evaluate K .

Rejections: none. The two interval-empty branches are unreachable: μ_2 empty needs $\bar{n} < 10$, and μ_1 empty needs $\bar{n} \leq 20$ or $\mu_2 \geq \bar{n}/10$, both excluded. The worst cv at the top of the feasible ω range is 0.184.

Matérn II

Primaries at λ_p , thinned by a hard core of radius R : a point survives if it carries the smallest mark within R . One dial, the hard core $c = R\sqrt{\bar{n}}$ in mean-spacing units, drawn log-uniform. This is an inhibitory family, so it has no smearing dial — there is no cluster to smear.

- $c \geq 0.05$ (`core_min`) — the floor.
- $\pi R^2 \bar{n} \leq 0.95$ (`matern_fill`) — the *packing* bound, entering as $c \leq \sqrt{0.95/\pi} = 0.5498$. Type II thinning cannot pass a fill of 1: writing $x = \pi R^2 \bar{n}$, the required primary intensity is $\lambda_p = \bar{n} [-\log(1-x)]/x$, which diverges as $x \rightarrow 1$. Capping at 0.95 caps the thinning ratio at $\lambda_p/\bar{n} = 3.145$.

λ_p is an *exact* inversion, not an approximation: x depends only on c , so $\lambda_p a = -\log(1-x)$ with $a = \pi R^2$, and the closed-form intensity $[1 - e^{-\lambda_p a}]/a$ returns $\bar{n}$ identically. Validated against the sampler across the fill range, realised $\bar{n}$ matches target to $\leq 0.4\%$.

Rejections: none, and the cv bound is unreachable by construction. A Matérn II process is under-dispersed relative to CSR, so over the whole design $cv \leq 1/\sqrt{\bar{n}} \leq 1/\sqrt{75} = 0.1155$, comfortably below $cv_{\max}$. The count-variance bound cannot bind for any inhibitory family at these $\bar{n}$. Realised cv runs 0.017–0.115.

LGCP

$\Lambda = e^{\mu_{\log} + Y}$ with $\text{Cov } Y = \sigma^2 e^{-r/s}$, simulated by circulant embedding on a $(2M)^2$ torus. Shape s , amplitude σ^2 ; $\mu_{\log} = \log \bar{n} - \sigma^2/2$ so the budget holds.

- $s \geq 0.10/\sqrt{\bar{n}}$ (`lgcp_s_min`) — the correlation range must be resolvable at the mean spacing. Below it the field decorrelates within a typical nearest-neighbour distance and the pattern is Poisson at every observable scale.

- $s \leq 0.20$ (`lgcp_s_max`) — absolute, and the largest range the circulant embedding supports with a non-negative spectrum.
- $\sigma^2 \geq 0.05$ (`lgcp_var_min`) — the amplitude floor.
- σ^2 capped by `cv_cap` at `cv_max`. This is the one place where the count-variance bound is the *active* ceiling for every draw: the nominal hard ceiling of 10 is never reached (realised σ^2 tops out at 8.82), and realised `cv` sits at exactly 0.300.
- Grid size M from `lgcp_grid.grid_size`: the smallest power of two whose discretisation bias in K is below 0.1 null s.d.s on the test’s radius set, searched over $[128, 4096]$. This is the one place the bank consults the null tables, and M is written to the manifest so the pattern stays reproducible from it.

Rejections: 865 of 50 865 **draws**, 1.70% — the only family that rejects at all. They are entirely the grid search failing to reach the tolerance within $M \leq 4096$: the s -interval-empty branch needs $\bar{n} \leq 0.25$, and the `cv_cap` failure branch needs `cv` at the σ^2 floor to exceed 0.30, whose maximum over the whole $(\bar{n}, s)$ domain is 0.143. The rejection is a genuine numerical infeasibility rather than a design bound, and it concentrates at the long-range end: 76% of accepted thetas need only $M = 128$, while 875 sit at the $M = 4096$ ceiling.

6 Pattern-level conditioning

Patterns are redrawn until $n \in [20, 2000]$, the null tables’ domain, so that every cloud in the bank can be given a statistic at all. This is conditioning, and it biases the pattern given θ — but only where it fires, and with the count-variance bound in place it essentially never does. Across all 500 000 patterns it fired **once**, for a single nested pattern. The realised n spans 20–1716.

7 Seeding

One addressable stream per (set, family, index, role):

```
Generator(PCG64DXSM(SeedSequence(root, spawn_key=(DV, set, family, index, role))))
```

with `role` separating the parameter draw from the pattern draw. Any stream can be rebuilt alone, so θ_i never depends on $\theta_{j < i}$: the bank can be extended, or a single cloud audited, without disturbing what exists. Shards are written atomically and skipped if present, so generation is resumable.

8 What the bank delivers

Family	thetas	rejected	$\tilde{\delta}$ (ext reduction)				$\tilde{\delta} < 1$
			q_{05}	q_{50}	q_{95}	max	
Poisson	50 000	0	−0.979	−0.972	−0.91	−0.88	100.0%
Thomas	50 000	0	−0.887	0.600	13.76	58.6	54.7%
Nested	50 000	0	−0.272	4.839	31.11	92.9	19.6%
Matérn II	50 000	0	−0.216	2.546	12.43	19.9	29.1%
LGCP	50 000	865	−0.847	0.860	20.73	96.7	51.8%

$\tilde{\delta} = 1$ is the 5% rejection boundary of the corresponding test, so the last column is the share of each family that is *not* detectable at $\alpha = 0.05$ from its L -function alone. Every family straddles it, which is the point: a bank that sat entirely on one side would measure nothing.

The zero of the scale is the noise floor, not CSR. `ext` takes the extremum of $\hat{L}(r) - r$ before studentising, and the extremum of a wiggly curve is displaced from 0 under CSR too. The reference

is therefore that free displacement, and $\tilde{\delta} = 0$ means “this departure equals CSR’s own noise floor”. Processes below it score negative — Poisson at -0.97 , and the q_{05} column for every family — and are deliberately left unclamped, so the label stays *ordered* below the detection boundary instead of piling up at a floor. `DELTA_EDGES` in `scripts/regimes.py` is geometric and does not reach there; that is a presentation choice still to be made, and `regimes.py` now reports the share of each family it excludes. The manifests carry four coordinates on the same bank: `delta_tilde` (the two-arm `ext`), its separate repulsion and clustering arms `delta_tilde_lo` and `delta_tilde_hi`, and the pointwise `delta_tilde_sup`. Each is calibrated so that 1 is its own boundary; they are not interchangeable as numbers. The two arms are a *signed* axis — repulsion and clustering are opposite departures that a single magnitude collapses together — and each arm sits at the noise floor across every family the other one carries.

Why the reduction was changed, and what it moved. The pointwise `sup` carries a radius window $[r_{\min}(\bar{n}), 0.25]$ with $r_{\min}(\bar{n})$ set at 5 expected pairs, and 32.6% of the Matérn bank has $R < r_{\min}(\bar{n})$ — the entire hard core below the window. The window is not optional for a *pointwise* statistic: $s_0(r) \rightarrow 0$ at small r , where CSR patterns rarely hold any pair, so the studentised ratio is unstable there. But a hard core’s detection boundary sits at $\bar{n}R \approx 1.6$, i.e. at $R/r_{\min}(\bar{n}) \approx 0.8$ — just inside the radii the window discards — so the `sup` draws its line slightly above where hard cores actually become detectable. Taking the extremum *before* studentising removes the need for a window entirely, because nothing is divided by a vanishing $s_0(r)$. It also makes the statistic exact rather than inferential on this family: below R no point has a neighbour, so $\hat{K} = 0$ and $\hat{L}(r) - r = -r$ identically, and $\min_r(\hat{L}(r) - r) = -R$ *noiselessly*. Measured power on matched n bears this out: at $\bar{n} = 200$ with core 0.1 ($R/r_{\min}(\bar{n}) = 0.79$) the `sup` rejects 19.3% of the time and the `lo` arm 81.0%; at $\bar{n} = 75$ with core 0.2, 65.4% against 100%. All four reductions hold exact size — 0.0501 overall for `ext` on 50 000 fresh off-grid patterns, within 2 s.e. in all six n bands.

The effect on the bank is a recalibration, not a reordering: `sup` and `ext` agree at $\rho = 0.999$ over the whole Matérn bank and 0.987 on the sub- $r_{\min}(\bar{n})$ stratum. What moves is *where the boundary falls*. On that stratum the share labelled detectable rises from 1.25% to 11.4%, and the median $\tilde{\delta}$ from 0.309 to 0.352; bank-wide the detectable share rises from 67.2% to 70.9% and the top of the range stretches from 13.0 to 19.9.

One caveat on what $\tilde{\delta}$ can order here. On the sub- $r_{\min}(\bar{n})$ stratum $\tilde{\delta}$ correlates with R alone at only $\rho = 0.17$ (and the `sup` at 0.08), but with $\bar{n}R$ at $\rho = 0.98$. That is correct rather than a failure: detectability depends on the core measured in units of the mean spacing, not on R , and $\bar{n}$ varies across the bank. A reduction scoring highly against R alone would be one that had stopped tracking detectability.

This is a property of the test, not a defect of the design: the bank’s job is to span the physical parameter, and it does. Forcing $R \geq r_{\min}(\bar{n})$ would require an $\bar{n}$ -dependent core floor and would delete exactly the near-CSR end of the Matérn design that the boundary analysis needs.

A resolution limit remains, and it is a property of the radius grid rather than of any reduction. `RADII` has step 0.25/512, so at $n \gtrsim 500$ a core at the detection boundary ($R \approx 1.6/n$) spans only one to three grid steps, and `lo` loses its advantage over the `sup` there.